EECS 498/CSE 598: Zero-Knowledge Proofs
Winter 2026
Lecture 11

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Agenda for this lecture

- Announcements
- The random oracle model
- The Fiat-Shamir transform (and related)
- Security of Fiat-Shamir
- Example of Fiat-Shamir for Schnorr

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Announcements

- Project 2 is online
- Autograder is working
- Due date: 2/20
- Anora

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Random oracle model

- Bellare – Rogaway '93
- Every party has access to global rand. func
  - ↳ P + V access <u>same</u> func
- <u>Lazy</u> sample RO entries

RO methodology

- Design scheme w/ oracle access to RO
- Prove security in RO model
- instantiate RO with crypto. hash
  - ↳ Difficult

SHA-256, SHA-3

Random oracle model

Uninstantiability

indifferentiability

CGH '98

Rothblum ...

Security in the ROM

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Fiat-Shamir: replace verifier with RO

P^f $\xrightarrow{R}$ V^f → public-coin

$c \leftarrow \mathbb{F}_p$

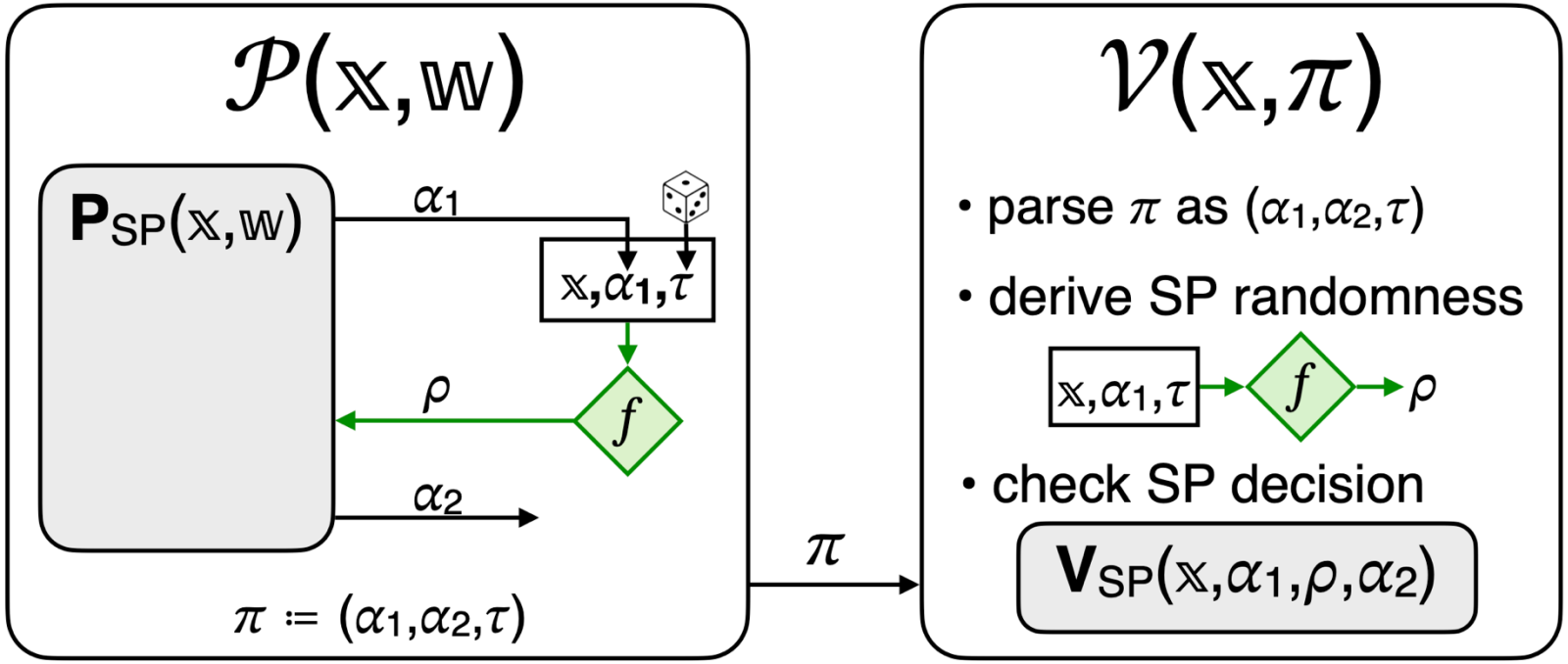
$\xleftarrow{c}$

$\xrightarrow{z}$

Query transcript up to V message

$F(\text{public input}, R)$

Fiat-Shamir: replace verifier with RO



(Courtesy Chiesa-Yogev)

Other transforms

Fischlin/Pass: straightline extraction

Unruh: QROM

"Quantum" ROM?

Query superposition of inputs,

get superpos. of outputs

"BCS" transform: multi-round

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Security of the Fiat-Shamir transform

”Weak” Fiat-Shamir attacks

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Making Schnorr non-interactive via F-S

$\mathbb{P}^{F}(\mathbb{G}, G, p, X; x)$:

$r \xleftarrow{\$} \mathbb{Z}_p$; $R \leftarrow rG$

$c \leftarrow F(\mathbb{G}, G, p, X, R)$

$z \leftarrow r + cx$

$\xrightarrow{R, c, z}$

$\mathbb{V}^{F}(\mathbb{G}, G, p, X)$:

$c' \leftarrow F(\mathbb{G}, G, p, X, R)$

If $c \neq c'$ Ret $\perp$

Ret $\boxed{zG = R + cX}$